

False Consensus: Why Intermediate Agreement Is Not a Safe Early-Exit Signal for Reasoning LLMs

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Abstract

Reasoning language models spend most of their tokens *after* the answer is effectively decided, so a popular way to cut inference cost is to periodically probe the partial trajectory for its current answer and stop once probes *agree*. We show this bet is unsafe, and that the failure is a property of the *signal*, not of the stopping threshold. On frozen reasoning trajectories from two models and three math benchmarks we run a *preregistered* search over a seven-dimensional consensus-rule schema (17,712 rules) with problem-grouped splits and acceptance gates fixed in advance. The result is distribution-free: **every one of the 17,712 rules loses accuracy in the worst case**—none reaches break-even, and stopping strictly lowers accuracy in 67.7% of rule–environment evaluations. The reason is directional, not statistical: intermediate consensus forms *while the answer is still moving*, so a stop destroys a correct-in-the-end answer roughly **35 times more often** than it rescues a wrong one—an *accuracy tax* that is a property of the trajectory, not the probe, and that no threshold in the searched space escapes. The same frontier reproduces on a held-out test split ($r=0.96$) and on unseen models at $4\times$ the scale and a different architecture, with no rule safe on both splits; a faithful Certaindex reproduction on identical trajectories collapses by 56–70 pp. The failure is specific to *consensus*: a method reading a different signal—the model’s own confidence at reasoning-boundary tokens—stays near full accuracy on the same trajectories, which localizes the problem to probe agreement and indicates where a safe early-exit signal must instead come from. We release the protocol, the frozen trajectories, and the full sweep.

1 Introduction

Reasoning language models such as DeepSeek-R1-Distill-Qwen-7B and Qwen3-8B (DeepSeek-AI, 2025; Qwen Team, 2025; OpenAI, 2024) solve

hard problems by emitting long chains of thought, often thousands of tokens per query. Much of this computation is spent *after* the model has effectively decided on an answer, which makes *early exit*—halting generation once the answer is settled—an appealing lever for cutting inference cost (Chen et al., 2024; Fu et al., 2024). A widely used family of methods does this by *probing*: at regular intervals the partial trajectory is queried for its current answer (e.g., by appending a short “Final Answer:” suffix), and generation is halted once several probes *agree* (Fu et al., 2024). The bet is that *intermediate consensus signals termination*—once the model repeatedly says the answer is x , further reasoning is unlikely to change it. This paper asks a single question: **is intermediate agreement a safe signal on which to stop?**¹

Our answer is **no**, and the reason is not a badly tuned threshold—it is the signal itself. Intermediate consensus forms *while the answer is still moving*: on a frozen replay, stopping on consensus destroys a correct-in-the-end answer roughly **35 times more often** than it rescues a wrong one (Section 5). This is an *accuracy tax* that is a property of the reasoning trajectory, not of how one probes it, so no threshold on the agreement signal can escape it. We make the claim hard to dismiss as “you did not tune it well” by searching the space exhaustively under commitment: on frozen trajectories from two models and three benchmarks we run a *preregistered* sweep of a seven-dimensional consensus-rule schema (17,712 rules) with problem-grouped splits and acceptance gates fixed in advance (Sections 3–4). The outcome does not depend on any single fragile estimate: **every one of the 17,712 rules loses accuracy in the worst case; none reaches**

¹We use “false consensus” for the phenomenon that intermediate probe agreement is non-terminal; the term is unrelated to the social-psychology “false consensus effect” (Ross et al., 1977). We study answer-level early exit (stopping a chain of thought), not token- or layer-level early exit (Graves, 2016; Schuster et al., 2022).

break-even. The frontier reproduces on a held-out test split ($r=0.96$) and on unseen models at $4\times$ the scale and a different architecture (Section 4), and a faithful Certainindex reproduction on the same trajectories collapses by 56–70 pp. Figure 1 is the whole paper in one picture: the safe-and-saving corner is empty for every consensus rule, the published consensus method sits at the far edge of the collapse, and the one method that stays near full accuracy is the one that does *not* read probe agreement.

Contribution. Our contribution is a rigorous, mechanism-backed demonstration that *intermediate answer-consensus is an unsafe early-exit signal for reasoning LLMs*, established so that it cannot be waved away as an untuned heuristic. Three components support this one claim:

- **A preregistered, probe-independent test** (frozen trajectories, offline re-probing, a seven-dimensional 17,712-rule schema, and selection gates fixed in advance) under which *no* rule is both safe and token-saving—and which we release in full (Sections 3–4).
- **A mechanism** that explains *why*: a directional “continued-reasoning-corrects” effect ($\approx 35:1$) makes the accuracy cost of stopping intrinsic to the trajectory and independent of probe density, so the negative result generalizes beyond the searched schema (Section 5).
- **Out-of-distribution and real-method confirmation:** the frontier reproduces on a held-out split and two unseen models, a published Certainindex reproduction collapses, and a non-consensus signal stays near full accuracy on the same trajectories—localizing the failure to consensus and indicating where a safe signal must come from (Sections 4–5).

2 Related Work

Probe-based / consensus early exit. The methods we scrutinize periodically query a partial reasoning trajectory for its current answer and halt on agreement. Dynasor / Certainindex (Fu et al., 2024) formalize this with a “certainty index” over recent probe answers and use it to terminate early when serving reasoning programs, one of several responses to the *overthinking* of reasoning models (Chen et al., 2024; Yang et al., 2025). We adopt the same probe mechanism (a short boxed-answer suffix) and notion of probe agreement, and ask how

far *any* rule in this family can be pushed before it becomes unsafe.

Early-stopping self-consistency and self-correction. Probe-based early exit is the online, *single-trajectory* analogue of early-stopping self-consistency (Wang et al., 2023; Aggarwal et al., 2023; Li et al., 2024): it votes across snapshots of one in-progress trajectory rather than across finished samples. That difference is the crux—snapshots of an unfinished trajectory are not exchangeable draws from a stable answer distribution, because the trajectory is still correcting itself. This intrinsic self-correction differs from the prompted, post-hoc self-correction that Huang et al. (2024) find unreliable: we measure answer evolution *within* one uninterrupted trajectory, and find (Section 4) that a model’s mid-trajectory answer is far from final.

Alternative termination signals. Rather than probe agreement, a stop can read the model’s own forward-looking state: verifier- and process-reward-guided decoding (Lightman et al., 2024; Wang et al., 2024) score partial reasoning, and adaptive-computation methods learn a halting signal at the token or layer level (Graves, 2016; Schuster et al., 2022). DEER (Yang et al., 2025) stops on the model’s confidence in a trial answer at reasoning boundaries. Our results (Section 5) indicate that this class—forward-looking signals about whether the trajectory will still change—is the direction a safe stop must take, in contrast to the backward-looking agreement signal we show fails. Test-time scaling trades compute for accuracy via repeated sampling and search (Snell et al., 2024; Brown et al., 2024); early exit is its dual, spending less without losing accuracy.

Preregistration. To keep an exhaustive rule search from silently overfitting to a favorable operating point, we fix the rule schema, splits, and acceptance gates before looking at held-out data (Nosek et al., 2025). To our knowledge this is among the first preregistered evaluations of early-exit rules for reasoning models; the protocol (Section 3) is what makes a *negative* result interpretable.

3 Method

Section 2 refuted one heuristic. To ask whether *any* probe-based rule can succeed, we need (i) an evaluation that a rule cannot game by changing the

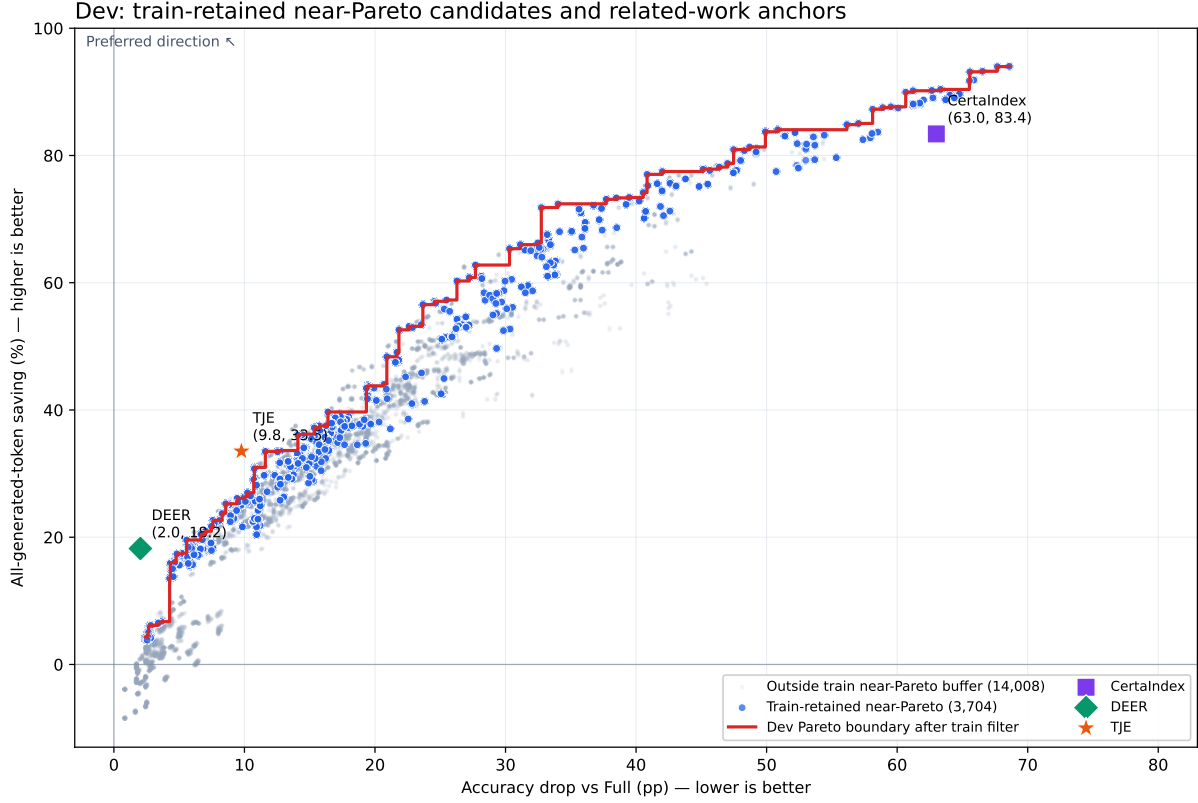


Figure 1: **The safe-and-saving corner is empty.** Each grey point is one of the 17,712 preregistered consensus rules on the development set; blue are the 3,704 surviving a near-Pareto train filter, and the red step is the dev Pareto boundary. The top-left corner (small accuracy drop *and* large token saving) contains no rule—the boundary only climbs as the drop grows. **CertaIndex** (Fu et al., 2024), the published consensus method, sits at the far right (63 pp drop), the consensus family’s collapse. **DEER** (Yang et al., 2025), which reads a boundary-confidence signal *outside* the swept space, occupies the low-drop corner (2.0 pp, 18%), above the consensus boundary at the same accuracy drop. The failure is a property of the consensus signal, not of early exit.

reasoning it observes, and (ii) a selection procedure that cannot silently overfit to a lucky operating point. We preregister both.

3.1 Probe-independent, frozen trajectories

For each problem we generate exactly *one* main trajectory in a single request (caps: 16K tokens for MATH500 and AMC23, 32K for AIME24) and freeze it. All probing is then performed *offline* against fixed text prefixes of that frozen trajectory. Because probes never re-enter the decode, changing a probe schedule—or a stopping rule—cannot change the underlying reasoning; every rule is scored against the same trajectories. We build two offline probe banks: a *dense* bank that re-probes every 64 tokens with simple@32 (a 32-sample boxed-answer probe), and an *adaptive* bank that additionally fires event-triggered probes at entropy drops and at conclusion/reflection/answer markers.

3.2 Token accounting

For a rule that stops a problem at token position s after consuming probe output p , we charge total decode tokens $T = s + p$ and compare against the frozen baseline B (the full trajectory, capped at budget). *Net savings* is $(B - T)/B$ and *gross savings* is $(B - s)/B$; their gap is the probe cost. The accuracy drop compares the correctness of the committed answer at s against that of the frozen final answer (it is not an agreement rate between them). Two quantities therefore move independently, and we track both: *where* the rule stops (setting the accuracy drop and the main-token saving s) and *how much it probes* to get there (adding p). Section 5 shows that only the first is intrinsic.

3.3 A seven-dimensional rule schema

A stopping rule is a point in a schema with seven independent dimensions: (1) **probe schedule** (fixed interval, or adaptive event triggers); (2)

Operating point	Max model drop	Max bench drop	psf floor
conservative	1.5 pp	2.0 pp	0.80
balanced	2.5 pp	3.0 pp	0.80
token_efficient	4.0 pp	5.0 pp	0.70

Table 1: Preregistered acceptance gates. “Max model/bench drop” cap the worst per-model and per-benchmark accuracy loss; psf is the fraction of environments with strictly positive net savings. Gates are fixed in advance and *not* relaxed post hoc.

maturity (minimum tokens or budget fraction before any stop is allowed, incl. an online-instability floor); **(3) evidence** (window share vs. latest-persistence vs. entropy-budget-fraction); **(4) persistence** (minimum consistent accepts and consensus span); **(5) certainty** (minimum fraction of supporting probes flagged certain); **(6) validity** (answer-shape filter, e.g., rejecting empty/letter answers); and **(7) history** (maximum answer switches, minimum stable span). Enumerating the preregistered grid over these dimensions yields 17,712 candidate rules spanning four families (window_share, latest_persistence, entropy_budget_fraction, adaptive_event_probe); formal definitions of each dimension are in Appendix A.

3.4 Models, benchmarks, splits

Development uses two models—DeepSeek-R1-Distill-Qwen-7B and Qwen3-8B—on three benchmarks (MATH500, AMC23, AIME24), each at seeds 42/43/44: 18 model×benchmark×seed environments. Problems are partitioned *by problem id* into 60/20/20 train/dev/test splits (MATH500 300/100/100, AMC23 24/8/8, AIME24 18/6/6), so no problem leaks across splits. Confirmation reserves seeds 45/46/47 plus two unseen models—DeepSeek-R1-Distill-Llama-8B and Distill-Qwen-32B—all evaluated on the *test* split only.

3.5 Preregistered operating points and gates

We fix three operating points and their acceptance gates *before* looking at held-out data (Table 1). Selection maximizes worst-case token savings subject to worst-case accuracy-drop caps and a positive-savings-fraction (psf) floor. A rule is accepted at an operating point only if it satisfies that point’s gate on train+dev.

3.6 Metric resolution

The *per-model* drop pools all of a model’s dev instances (≈ 114 problems per seed, 342 over three seeds), so its resolution is fine (~ 0.3 pp) and the 1.5 pp conservative cap is meaningful. The *per-benchmark* drop is coarser on the small competition sets: one AIME24 problem is 16.7 pp of a 6-problem dev split (5.6 pp pooled over three seeds) and one AMC23 problem is 12.5 pp (4.2 pp pooled), so a 2.0 pp per-benchmark cap is finer than a single problem can resolve there. We therefore treat the **per-model gate as primary** and the per-benchmark gate as a *diagnostic*, and we report the accuracy floor of Section 4 as a per-model (pooled) quantity that is not driven by single-problem quantization. Confidence intervals for the frontier floor use a stratified bootstrap over the nine benchmark×seed environments (Section 4); held-out confirmation provides the complementary out-of-distribution check.

3.7 Preregistration commitments

Three commitments make the result interpretable. **(C1)** The test split and both confirmation models are never read during rule selection. **(C2)** The conservative gate in Table 1 is not relaxed after seeing results; if no rule passes, that is reported as the finding rather than engineered away. The balanced and token_efficient points are reported only as clearly-labeled post-hoc analysis, never as a selected operating point. **(C3)** The frozen rule set is fixed on train+dev and its hash recorded before any test/confirmation evaluation. Section 4 reports what happened when we applied C2.

4 Experiments

We first confirm the phenomenon the paper turns on—intermediate consensus is non-terminal (§4.1)—then report the preregistered sweep and its central negative result (§4.2), reproduce three prior early-stoppers on identical trajectories (§4.3), and confirm the frontier out of distribution (§4.4).

4.1 False consensus is real

We log a dense probe stream for every MATH500 problem under DeepSeek-R1-Distill-Qwen-7B, never intervening—the probe is a separate forward pass that only observes the main trajectory (full setup in Appendix E)—and read off four facts.

(1) Agreement is not correctness. *Cumulative* agreement (all non-empty probes so far concur) is

nearly trustworthy (98.9% correct on the 87 problems reaching it), but that is not what an online rule sees. The operative signal, *window* agreement (the last five probes concurring), is markedly less reliable: of the 338 problems reaching a unanimous last-five window, only 93.5% are correct—a 6.5% false-consensus rate. **(2) Recovery is common.** Of the 375 problems whose *first* probe is wrong, 76.3% eventually flip to correct; of the 145 that formed a three-probe consensus *different* from their final answer, 65.5% ended up correct. **(3) Later consensus is worse.** Consensus formed before 512 tokens is 87.4% correct, but consensus forming only after 2048 tokens is 58.1% correct—hard problems converge slowly *and* to worse answers. **(4) The cost lands at the stop.** A naive stop (first three consecutive agreeing, certain, non-empty probes) fires on 416/500 problems and commits to answers correct only 69.2% of the time, whereas letting those same problems run to completion reaches 85.6%: a 16.4 pp loss.²

A good stopping rule must therefore be much more than “agreement.” The rest of this section asks whether the *best possible* such rule, drawn from a large preregistered space, can be both safe and economical.

4.2 The safe-and-saving corner is empty

We aggregate the sweep into per-environment metrics—637,632 rows = 17,712 rules $\times$ 18 environments $\times$ 2 splits (train, dev)—and compute, per rule, its worst per-model and per-benchmark accuracy drop and its net token savings (accounting of Section 3.2). Applying commitment C2 to the preregistered conservative gate yields the paper’s central result, and it does not rest on any single fragile estimate.

The headline is distribution-free: *every* one of the 17,712 candidate rules incurs a positive worst-case per-model accuracy drop, and *zero* reach break-even. Across all 637,632 rule–environment evaluations, accuracy strictly *falls* in 67.7% and strictly *rises* in only 6.7% (unchanged in 25.5%), for a mean drop of 10.9 pp; over rules the worst-case per-model drop has percentiles $p_1=3.37$, $p_5=4.26$, $p_{25}=10.7$, median 20.1 pp. The Pareto frontier over (worst-case drop, net savings) con-

tains 93 rules and **none clears the conservative gate** (per-model drop ≤ 1.5 pp, per-benchmark ≤ 2.0 pp, $\text{psf} \geq 0.8$). Figure 1 shows the shape: the boundary only climbs as the accuracy drop grows, and the small-drop / large-saving corner is empty. Two claims should be kept separate: the safety claim (no rule reaches even 1.5 pp; all lose) is about accuracy and, as Section 5 argues, is probe-independent; the savings claim (positive net savings costs ≥ 4.87 pp, whether the psf floor is 0.8/0.7/0.6/0.5) additionally depends on the dense-probe accounting.

On the least-bad rule (and why we do not lean on it). The smallest per-model drop anywhere is 1.85 pp—already above the gate—from an `entropy_budget_fraction` rule that stops so rarely its net savings are negative ($q_{20} \approx -11\%$). We report it as a diagnostic, not a load-bearing number: a bootstrap over the nine benchmark $\times$ seed environments puts it at 1.85 pp with a 95% CI of [0.0, 5.6] pp, touching break-even. We rest the result instead on the two facts that do not depend on this point estimate—all 17,712 rules lose accuracy, and even this least-bad rule spends *extra* tokens (net -8 to -9%)—and on the held-out reproduction of §4.4. The per-model gate is primary; the per-benchmark cap is a diagnostic a single competition-set problem cannot resolve (Section 3.6).

Sophistication does not help. The ordering across families is one-directional: the simplest evidence signal (`window_share`) gives the best safe-with-positive-savings point (4.87 pp); `entropy_budget_fraction` reaches 1.85 pp only by stopping so rarely savings go negative; and `adaptive_event_probe`—the most sophisticated family—is *dominated*, needing 9.70 pp for a positive-savings point (frontiers in Appendix D). Table 2 places our best swept points against full generation; none passes the gate, and even the accuracy-neutral point spends $\sim 4\%$ extra tokens (probe tax, Section 5).

4.3 Prior stoppers on the same trajectories

To place published methods on the same footing, we reproduce three in a common harness—the same frozen trajectories, the same simple@32 probe bank, the same token accounting. **CertainIndex** (Fu et al., 2024) is the faithful share-threshold stop (the direct descendant of the swept family); **DEER** (Yang et al., 2025) stops on the model’s own confidence in a trial answer at reasoning

²This illustration uses a tight 3072-token budget so truncation is visible on a single A100 and a simplified heuristic, *not* the CertainIndex algorithm (§4.3). The main study below runs at the 16K/32K budgets used throughout, where the same effect appears as the directional accuracy tax of Section 5.

Method	Accuracy	Δacc (macro)	Net token saving
Full generation	82.7%	—	0%
Ours (target conservative)	81.8%	−0.9 pp	−4.0%
Ours (target balanced)	79.1%	−3.6 pp	+7.8%
Ours (target token_efficient)	77.1%	−5.6 pp	+19.6%

Table 2: Best swept operating points on dev (macro over 18 environments): the min-drop, mid-frontier, and high-saving representatives of the dev Pareto frontier. The target-conservative rule is roughly accuracy-neutral in macro but spends $\sim 4\%$ extra tokens (probe tax) and fails the per-model gate.

boundaries—a *different* signal; **TJE** (Anonymous, 2026) is a token-level stop. Table 3 reports the reproductions (not the original papers’ end-to-end runs).

CertaIndex, at 99.8% stop rate and 70 pp lost, is the accuracy tax paid in full; in Figure 1 it sits at the far right of the collapse, while DEER—outside the swept space—occupies the low-drop corner. TJE is unsafe on the weaker model. That only the non-consensus method survives is the crux we analyze in Section 5.

4.4 The frontier reproduces out of distribution

We read the test split once (commitment C3). Because the development result is that *no* rule clears the gate, there is nothing to confirm by re-selection; we ask whether the *frontier* reproduces, ranking rules on dev and only measuring on test.

Held-out test split. Across all 17,712 rules, each rule’s worst-case per-model drop on test tracks its dev value ($r = 0.96$), and every rule still loses. Toward the decision boundary the correlation is noise-limited on both sides ($r \approx 0$ for dev drop ≤ 5 pp; the 144 rules with dev drop ≤ 3 pp collapse to a single test value, +0.11 pp). The split-invariant claim is the empty joint gate: **no rule clears the conservative gate on both splits**—0 on dev, and of the 272 rules admissible on *test alone*, not one is admissible on dev (there they cost 4.98–5.65 pp, median 5.09). These 272 are exactly the in-sample winners the held-out split exists to reject.

Unseen scale ($4\times$) and architecture. On Distill-Qwen-32B, never used in development, the frontier reproduces ($r = 0.903$) and the dev-selected “safe” rules do not transfer: rules with dev drop ≤ 1.5 pp cost a mean +2.58 pp

and up to +8.33 pp on test, and the dev floor rule drops +8.33 pp there. Scaling $4\times$ opens no safe corner. The held-out architecture (DeepSeek-R1-Distill-Llama-8B) reproduces the same qualitative pattern.³ Across the held-out split and both unseen models, no rule is jointly safe-and-saving.

5 Analysis

Section 4 is an existence proof over 17,712 rules, two seen and two unseen models, and both splits. This section explains *why* it holds and why it generalizes beyond the exact rules we searched: the accuracy cost of stopping on consensus is a directional property of the reasoning trajectory itself, not of a threshold or of probe density.

5.1 Stopping destroys more than it saves: a 35:1 effect

Consider only the stops where committing early *changes* correctness relative to the trajectory’s own final answer, and count two outcomes: (final-correct, stop-wrong)—a correct recovery destroyed—versus (final-wrong, stop-correct)—a wrong answer luckily banked. Pure sampling noise predicts a $\approx 1:1$ ratio; a genuine “continued reasoning corrects” effect predicts $\gg 1$. On the frozen replay this ratio is **35:1** for the naive consensus stopper (1055 destroyed vs. 30 banked; 34.3:1 and 35.9:1 on DeepSeek-R1-Distill-Qwen-7B and Qwen3-8B separately) and 15–18:1 for the more conservative variants—all far above 1:1. Stopping on consensus removes a correct-in-the-end answer about 35 times more often than it rescues a wrong one. This is the recovery phenomenon of §4.1 (65.5%, 76.3%), now at the main 16K/32K budgets and as a signed effect: the loss is real and directional, not sampling variance.

5.2 The accuracy tax is intrinsic to the trajectory

Why is the ratio structural? For a rule stopping at token position s , the accuracy drop depends only on s : committing at s forgoes whatever correction the trajectory would have made after it. It is a property of the reasoning, not of the probe, so it survives *any* probing scheme. This is what never reaches zero in

³The Llama-8B trajectories were re-collected after correcting a chat-template defect (a missing model-specific beginning-of-sequence token that degraded its generations); the corrected model scores $\sim 88\%$ on MATH500. Full corrected numbers are in the released sweep.

Method	Model	Accuracy	Δacc	Fair token saving	Stop rate	Signal
Full generation	Qwen3-8B	85.4%	—	0%	0%	—
Full generation	DeepSeek-R1-Distill-Qwen-7B	79.8%	—	0%	0%	—
CertaIndex	Qwen3-8B	15.3%	−70.1 pp	90.1%	99.8%	consensus
CertaIndex	DeepSeek-R1-Distill-Qwen-7B	23.9%	−55.9 pp	76.7%	98.7%	consensus
TJE	Qwen3-8B	85.0%	−0.4 pp	2.0%	22.5%	token-level
TJE	DeepSeek-R1-Distill-Qwen-7B	60.7%	−19.1 pp	65.0%	93.4%	token-level
DEER	Qwen3-8B	86.2%	+0.8 pp	16.3%	41.4%	boundary conf.
DEER	DeepSeek-R1-Distill-Qwen-7B	74.9%	−4.8 pp	20.2%	56.1%	boundary conf.

Table 3: Prior early-stoppers reproduced on our frozen trajectories (dev, macro over three benchmarks; probe tokens charged). CertaIndex—the consensus descendant—**collapses** (−56 to −70 pp) while saving 77–90% of tokens: it fires on almost every problem, very early, exactly the false-consensus trap of §4.1. **DEER**, which reads a *different* signal (boundary confidence), is the only method that stays near full accuracy—+0.8 pp on Qwen3-8B—while still saving tokens. We interpret this contrast in Section 5.

the sweep. The probe grid (intervals 64/128/256 tokens) makes essentially every stop position reachable and the observed floor is the minimum over these schedules; a sparser schedule can only stop later or at a subset of positions, never at a safer one the grid missed. Hence no threshold on the consensus signal—in or beyond our schema—escapes the accuracy tax; the searched space is a lower bound on the cost, not a lucky miss.

5.3 Ruling out the “it is just dense probing” objection

The natural rebuttal is that the negative result is an artifact of charging dense re-probing. It is not, and separating two costs shows why. Besides the accuracy tax, a rule pays a *probe tax*: the probe output p (Section 3.2). Under dense re-probing a late-stopping rule pays p over a long prefix, which turns the *net* savings of the safe (entropy) family negative even though its *gross* savings is positive (Table 4). Unlike the accuracy tax, the probe tax scales with probe density and is reducible (sparser or shorter probes, KV-cache reuse). The two are separable: the *savings* collapse is partly a probe-tax artifact and is addressable, but the *accuracy* floor is probe-independent and is what makes the corner empty.

5.4 It is the signal, not early exit

The reproductions of §4.3 localize the cause. Every method that stops on *probe agreement*—the swept schema and CertaIndex alike—either fails the safety bar or collapses outright; CertaIndex’s 70 pp loss at 99.8% stop rate is the accuracy tax paid in full. The lone method that stays near full accuracy, DEER, does not read probe agreement at all: it stops on the model’s own confidence in

Operating point	Gross saving $(B - s)/B$	Net saving $(B - s - p)/B$
Ours (target conservative)	+0.6%	−4.0%
Ours (target balanced)	+10.0%	+7.8%
Ours (target token_efficient)	+21.5%	+19.6%

Table 4: Gross vs. net savings; the gap is the probe tax. The target-conservative point’s negative net saving is a probing-density artifact, not evidence that stopping wastes tokens—whereas the accuracy tax above it is intrinsic.

a trial answer at a reasoning boundary, and in Figure 1 it sits *above* the consensus Pareto boundary at the same accuracy drop. This is direct evidence that the failure is a property of the *consensus signal*, not of early exit. (DEER’s edge is model-dependent, +0.8 pp on Qwen3-8B vs. −4.8 pp on DeepSeek-R1-Distill-Qwen-7B, and these are frozen-trajectory reproductions.)

5.5 Implication: change the signal, not the threshold

Much prior effort tunes consensus-based stops (Fu et al., 2024; Aggarwal et al., 2023; Li et al., 2024); our frontier shows the ceiling of that approach is low, and the mechanism above says no amount of tuning raises it. The constructive implication is that a safe signal should be about whether the *trajectory itself* will still change, not about the agreement of backward-looking probes. As a first step, we sketch in Appendix F an online controller built on DEER’s boundary-confidence signal (fast-commit on high confidence, else a verification branch) that recovers substantial savings at near-neutral accuracy on dev; we stress it is exploratory (three seeds, seed-sensitive, outside the preregistration) and report it

only to show that stepping outside the consensus-signal space escapes the accuracy tax we proved unavoidable inside it. Verifier- and process-reward-based termination (Lightman et al., 2024; Wang et al., 2024) and learned halting (Graves, 2016; Schuster et al., 2022) point the same way. Turning any of these into a *confirmed* method—with its own train/dev→test validation—is the natural next paper; our contribution is to establish rigorously which signal not to build it on. Two caveats bound the claim: it concerns *consensus-based* rules within the searched schema and dense-probe regime, and it does not say early exit is impossible—only that this signal cannot make it safe.

6 Conclusion

We asked whether intermediate answer-consensus is a safe signal on which to early-exit a reasoning LLM. It is not, and not because the rule is untuned: intermediate consensus forms while the answer is still moving, so stopping on it destroys a correct-in-the-end answer about 35 times more often than it rescues a wrong one. This accuracy tax is a property of the trajectory rather than the probe, and it shows up as an empty safe-and-saving corner across a preregistered space of 17,712 rules—every one of which loses accuracy—reproduced on a held-out split and two unseen models, and paid in full by a faithful Certaindex reproduction that collapses by 56–70 pp. Because a method reading a non-consensus signal stays near full accuracy on the very same trajectories, the failure is specific to consensus. The takeaway for anyone building safe early exit is therefore about the choice of signal: read the model’s own forward-looking state, not the agreement of its backward-looking probes.

Limitations

Scope of the negative result. The central result—every rule loses and none clears the conservative gate—is established on 18 development environments (two models, three benchmarks, seeds 42/43/44) and confirmed on the held-out test split and two unseen models (Section 4). It is a statement about the space searched: one boxed-answer probe suffix (simple@32) and one preregistered schema (17,712 rules, four families, seven dimensions). Other probe wordings and rule dimensions outside this schema are not searched; the mechanism (Section 5) is our argument that they would not help, but it is an argument, not an exhaustive

proof.

The exact floor is a development quantity. The 1.85 pp least-bad-rule value has a bootstrap 95% CI of [0.0, 5.6] pp and measures 0.11 pp on test; we therefore do not rest any claim on it (Section 4.2). The load-bearing findings are the distribution-free ones—all 17,712 rules lose, the empty joint gate—which do not depend on the floor’s point estimate.

Held-out architecture was re-collected. The held-out architecture model (Llama-8B) was re-collected after we found and fixed a chat-template defect (a missing model-specific beginning-of-sequence token) that had degraded its earlier generations; the corrected model scores $\sim 88\%$ on MATH500, and the confirmation uses the corrected trajectories.

Small held-out sets on hard benchmarks. AIME24 and AMC23 are small (dev splits of 6 and 8 problems per seed), so a single problem is a large fraction and per-benchmark cells are noisy. We aggregate over seeds and treat the per-model gate as primary and the per-benchmark gate as a diagnostic (Section 3.6).

Probe-tax dependence of the savings axis. Net-savings numbers charge dense re-probing (every 64 tokens, 32-token probes); a sparser or KV-cache-reusing schedule would change the *savings* axis and could move some rules to positive net savings. We separate gross from net savings (Section 5) precisely so the probe-tax-dependent part is distinguishable; the accuracy-tax result does not depend on the schedule.

The constructive signal is exploratory. The boundary-confidence controller of Appendix F spans three dev seeds but is seed-sensitive (macro Δacc per-seed range $[-6.1, +4.2]$ pp; mean -0.75 pp), model-dependent, not part of the pre-registered sweep, and has no test-split confirmation. It should be read as evidence that a non-consensus signal escapes the accuracy tax, not as a finished method.

Domain. All benchmarks are competition mathematics with checkable final answers. Whether false consensus and the accuracy tax transfer to open-ended reasoning, code, or agentic tasks is out of scope.

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- ## A Rule Schema Details
- The seven-dimensional schema of Section 3 enumerates to 17,712 candidate rules. Each rule fixes:
- Probe schedule:** fixed interval $\in \{64, 128, 256\}$ tokens, or adaptive_event with entropy-drop and marker triggers and a fallback interval.
 - Maturity:** fixed_tokens (minimum tokens), budget_fraction (minimum fraction of the decode budget), or online_instability (a floor plus a same-answer stability check).
 - Evidence:** window share, latest persistence, or entropy-budget-fraction candidate selection.
 - Persistence:** minimum consistent accepts and minimum consensus span in tokens.

5. **Certainty:** minimum fraction of supporting probes flagged certain.
6. **Validity:** answer-shape filter (reject empty / single-letter answers to suppress the Type-E probe artifact of Appendix E).
7. **History:** maximum answer switches within a window and minimum stable span since the last switch.

Certainty. A probe is *certain* when its boxed answer is emitted with high token-level confidence: concretely, the mean probability of the answer tokens under the probe forward pass exceeds a fixed threshold (the Certaindex certainty flag). For the simple@32 bank this is aggregated over the 32 samples as the fraction agreeing on the modal answer. Dimension 5 gates a stop on the fraction of the supporting probes that are certain.

Entropy triggers. The adaptive_event schedule fires an extra probe when the token-level predictive entropy of the main decode drops by at least ΔH relative to a trailing baseline (an “the model just committed” signal), and at conclusion/reflection/answer marker tokens (e.g., “therefore”, “the answer is”), subject to a minimum inter-probe gap and a fallback interval.

B Preregistration Text

The protocol version, split manifest hash, candidate-rule generation script, and the three operating-point gates (Table 1) were fixed before any held-out evaluation. Commitments C1–C3 (Section 3) were recorded with the protocol. When the conservative gate proved empty, we followed C2 and report the empty gate as the finding rather than relaxing it; the balanced/token_efficient points are reported only as clearly-labeled post-hoc analysis and would be re-preregistered if used for selection.

C Reproducibility

Frozen trajectories, offline probe banks, the sweep archive (637,632 metric rows), and the aggregation scripts are released. Table 6 was reproduced to the reported precision directly from the released sweep archive. Grading uses a robust answer-equivalence check that tries the raw and normalized reference forms and both string and math-expression equality; per-problem correctness for the frozen baseline

Rule family	Per-model drop (pp)	Net savings at that point
entropy_budget_fraction	1.85	negative ($q_{20} \approx -11\%$)
window_share	4.87	≈ 0 ($q_{20} \approx 0.3\%$)
latest_persistence	8.20	$q_{20} \approx 9.7\%$
adaptive_event_probe	9.70	$q_{20} \approx 4.5\%$

Table 5: Best operating point per rule family on dev. For entropy_budget_fraction we show its *absolute* minimum drop (which has no positive-savings point); for the other three families we show the minimum worst-case per-model drop *among positive-savings points*. The safest family cannot save; the most complex family (adaptive) is dominated by the simplest (window share).

Adaptive point	Trigger	Mean save	q_{20} save	Macro Δ acc
conservative	entropy drop	14.2%	4.5%	−5.88 pp
moderate	conclusion	25.9%	20.0%	−9.11 pp
aggressive	conclusion	37.3%	27.1%	−15.03 pp

Table 6: The adaptive_event_probe frontier on dev. Worst per-model/per-benchmark drops for the three points are −9.70/ − 10.42, −13.65/ − 12.50, and −19.28/ − 18.75 pp respectively. Reproduced exactly from the released sweep archive.

is recomputed at replay time rather than trusted from collection.

D Family and Adaptive Frontiers

Table 5 gives each family’s best attainable operating point on dev (supporting Section 4): the simplest evidence signal (window_share) gives the best safe-with-positive-savings point, the safest family (entropy_budget_fraction) cannot save, and the most complex family (adaptive) is dominated by the simplest. Table 6 isolates the adaptive frontier, showing it trades linearly into larger losses as it grows aggressive.

E False Consensus: Full Analysis

Section 2 summarizes the four facts this appendix establishes in full, in an intervention-free setting: that intermediate probe consensus is not a reliable stopping signal.

E.1 Setup

We run DeepSeek-R1-Distill-Qwen-7B (vLLM, one A100-80G) on all 500 MATH500 problems (Lightman et al., 2024; Hendrycks et al., 2021) with a decode budget of 3072 tokens, temperature 0.6, top- p 0.95, and seed 42. Every 128 generated tokens (at most 24 probes per problem) we append

the boxed-answer probe suffix and read off the model’s current answer. The probe is a separate forward pass: the controller only *observes*, never halting or altering the main trajectory. This yields, per problem, the full answer trajectory a stopping rule would have access to. This exploratory study uses a deliberately tight 3072 budget so that many trajectories are truncated and the “continue vs. stop” contrast is visible within one A100; the preregistered sweep (Section 3) re-examines the phenomenon at the 16K/32K budgets used for the main results, and we report the recovery statistics there as well.

E.2 Agreement is not correctness

We compare two notions of probe agreement. *Cumulative* agreement (all non-empty probes so far concur) is nearly trustworthy: on the 87 problems that reach unanimous cumulative agreement, 98.9% are correct, with a single whole-trajectory false consensus. But cumulative agreement is not what an online stopping rule sees. The operative signal is *window* agreement—the last five probes concurring—and it is markedly less reliable: of the 338 problems reaching a unanimous last-five window, only 93.5% are correct, giving a 6.5% (22/338) false-consensus rate. Perfect *global* consensus is almost safe; the *local* consensus a controller must act on is not.

E.3 The cost lands at the stopping decision

We simulate a *naive consecutive-agreement* stop: halt at the first point where three consecutive probes agree, are non-empty, and are *certain* (the model emits the boxed answer with high token-level confidence; formal definition in Appendix A). This is a simplified heuristic, *not* the Certaindex algorithm, which we reproduce faithfully in §4.3. It fires on 416/500 problems. The committed answers are correct only 69.2% of the time, whereas letting those same problems run to completion (within the 3072 budget) reaches 85.6%—a 16.4 **pp accuracy loss** in exchange for an average saving of 1321 tokens, with 128 problems stopped on an outright wrong answer.

E.4 Recovery is common, and early stopping kills it

The loss is driven by *recovery*: continued reasoning overturns the intermediate answer far more often than intuition suggests. Of the 145 problems that at some point formed a three-probe consensus *different* from their final answer, 95 (65.5%) ended up

correct; and of the 375 problems whose *first* probe was wrong, 76.3% eventually flipped to correct. Neither “first impression” nor “local consensus” is close to terminal.

Counterintuitively, *later* consensus is *less* trustworthy: consensus formed before 512 tokens is 87.4% correct, but consensus that only forms after 2048 tokens is 58.1% correct. Hard problems both converge slowly and converge to worse answers—exactly the problems on which an aggressive stopping rule does most damage.

E.5 Miscalibration

Window share is a poorly calibrated confidence estimate. Its calibration error—the gap between share and empirical accuracy, averaged over share bins (CCE)—is 0.080 (versus 0.149 for cumulative share), but the error is concentrated where a rule would act: at window share 0.8 the empirical accuracy is only 47.2% ($n=36$). A threshold tuned to “looks nearly unanimous” is operating precisely in the band where the signal is least reliable.

E.6 An error taxonomy

Hand-inspecting stopped-but-wrong cases (28 cases from the first 100 problems; AI-assisted labeling, treated as preliminary) yields *four* non-empty types out of five categories. By count: **(A)** numeric collapse—stable convergence to a wrong number—14/28; **(D)** derivation gaps—dropped roots/cases, unverified solutions—7/28; **(E)** format/option hallucination—a non-multiple-choice problem stably emitting a letter such as “B”—6/28, which is an artifact of the probe mechanism itself rather than a model belief; and **(B)** expression collapse—1/28. Type **(C)** (sign errors) did not occur in this sample. Type E is itself a methodological finding: roughly one in five “false consensus” is a probe formatting artifact, implying that any deployed rule should filter answer *shape*.

Takeaway. These observations—local consensus unreliable, recovery common, late consensus worse, confidence miscalibrated—suggest that a good stopping rule must be much more than “agreement.” The rest of the paper asks whether the *best possible* such rule, drawn from a large preregistered space, can be both safe and economical. In the searched space, it cannot.

F Exploratory: A Boundary-Confidence Controller

Section 4.3 showed that the one prior method staying near full accuracy, DEER, reads a different signal. As a first step beyond the main (negative) result, we push that signal into a deployment-style online controller and find it recovers substantial savings at near-neutral accuracy on dev. We stress up front that these results are *exploratory*—three dev seeds, outside the preregistered sweep, seed-sensitive, and without test-split confirmation—and we report them only to establish that the failure is about the *signal*, not early exit. They are not a validated method and are not part of the paper’s claims.

F.1 From trial-answer confidence to a verification branch

DEER emits a short trial answer at reasoning-boundary (“Wait”) tokens and stops once the model’s confidence in that trial answer (the mean probability of its answer tokens) exceeds a threshold. On our development trajectories the mechanism has a flaw: DEER measures confidence on the trial answer but then *delivers a separately generated readout*, and the two disagree on 14.6% of triggered problems. The extra readout averages 470 tokens yet yields no net accuracy gain (trial 88.9% vs. readout 88.5%), though it corrects some cases on some models. We therefore evaluate an *inspired* controller that (i) *fast-commits* the trial answer when confidence is very high (> 0.995), skipping the redundant readout, and (ii) otherwise opens a *retained verification branch* that commits only if a second high-confidence answer is mathematically equivalent, and otherwise keeps reasoning. Both run in a single online engine that generates the main trajectory live and probes only at scheduled boundaries—no frozen suffixes.

F.2 Online deployment results

Table 7 reports online runs over the 18-environment development set at three seeds (42/43/44; 1,368 trajectories). Across seeds the inspired controller holds macro accuracy near-neutral (-0.75 pp; per-seed range $[-6.1, +4.2]$ pp) while saving 34% of tokens (per-seed 25–44%), still dominating a faithful online DEER reference (-2.7 pp, 22%). A stratified paired bootstrap over the 18 environments (10,000 resamples) gives a token-saving advantage of $+12.1\%$ (95% CI $[+0.7, +22.9]$) at an accuracy

Method	Model	Δ acc	Fair saving
Inspired	Qwen3-8B	-1.5 pp	41.0%
Inspired	DeepSeek-7B	$+0.0$ pp	27.4%
Inspired	macro	-0.75 pp	34.2%
DEER ref.	macro	-2.7 pp	22.1%

Table 7: Online boundary-confidence early exit (dev, seeds 42/43/44, 1,368 trajectories; macro over 18 environments, per-model over three seeds). The inspired fast-path + verification-branch controller keeps macro accuracy near-neutral (-0.75 pp, per-seed range $[-6.1, +4.2]$) while saving 34% of tokens, and saves significantly more than a faithful online DEER reference at indistinguishable accuracy. Exploratory: seed-sensitive, small AMC23/AIME24 sets.

difference of $+2.0$ pp (95% CI $[-5.0, +9.0]$): the controller saves significantly more at statistically indistinguishable accuracy. The single-seed 42 point ($+2.8$ pp on Qwen3-8B, 44% saving) sits at the favorable end of this spread.

F.3 Relation to the preregistration

This controller is *not* part of the preregistered consensus sweep: it uses a different signal, a separate configuration, and its own results directory, and it would require its own train/dev→test validation before any confirmed claim. It does not modify the sweep’s candidates, gates, or selected rules (there were none). We therefore present it as a positive *signal*, not a finished method: it shows that stepping outside the consensus-signal space escapes the accuracy tax that Section 4 proved unavoidable inside it.